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Do good: Strategies for leading an inclusive data science or statistics consulting team

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Leading a data science or statistical consulting team in an academic environment can have many challenges, including institutional infrastructure, funding and technical expertise. Even in the most challenging environment, however, leading such a team with inclusive practices can be rewarding for the leader, the team members and collaborators. We describe nine leadership and management practices that are especially relevant to the dynamics of data science or statistics consulting teams and an academic environment: ensuring people get credit, making tacit knowledge explicit, establishing clear performance review processes, championing career development, empowering team members to work autonomously, learning from diverse experiences, supporting team members in navigating power dynamics, having difficult conversations and developing foundational management skills. Active engagement in these areas will help those who lead data science or statistics consulting groups – whether faculty or staff, regardless of title – create and support inclusive teams.

KEYWORDS
data science or statistical consulting team, inclusive team, leadership, management

1 | INTRODUCTION

What does it mean to support diversity, equity and inclusion as the leader of an academic data science or statistical consulting team? While discussing the many components of this large topic as part of a 2023 workshop on Models for Sustainable and Inclusive Data Science Consulting and Collaboration in Higher Education, we covered a number of areas that are undeniably important – from hiring practices to workforce and professional development. Yet in discussing the barriers to building diverse teams where the members are able to do their best work and thrive regardless of their personal characteristics or identities, we kept returning to examples of, as Robert Sutton puts it, "assholes" we had encountered (Sutton, 2007).

The asshole team leaders of our examples stole or failed to give credit to those on their team, took advantage of power differentials, refused to define performance review criteria, micromanaged, promoted underperforming friends and undermined others. While most of the behaviours we observed stopped short of the insults, threats, shaming and humiliation that Sutton (2007) details, they nevertheless had a similar effect of making team members feel disrespected and de-energized. These behaviours undermine efforts to develop an inclusive team, and many of them also preclude implementing best practices for reducing bias in hiring processes and the recruitment and retention of a diverse team.

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Supporting inclusion as a leader goes beyond just inviting diverse voices to the table. It requires actively promoting a culture that enables people to feel appreciated, valued, safe, respected, engaged and able to be authentic. This in turn facilitates their ability and willingness to contribute their best to their work, leading to organizational success (Ferdman et al., 2021). Here we focus on going beyond "doing no harm" and advocate for practices team leaders can use to increase diversity, equity and inclusion on statistics and data science consulting teams. These "doing good" practices generally fall under the umbrella of established best practices for management and leadership (Morgan et al., 2022). Unfortunately, most data science or statistics consulting team members in academic institutions, from students to centre directors, receive very limited leadership and management training (Leiserson & McVinney, 2015). The lack of such fundamental practices compromises diversity, equity and inclusion initiatives at all levels because they have no foundation on which to build and limited accountability.

There are many ways team leaders can support an inclusive environment, from flexible work arrangements to creating mechanisms for employee feedback. This paper highlights nine leadership and management practices that are especially relevant to the dynamics of data science or statistics consulting teams in an academic environment: ensuring people get credit, making tacit knowledge explicit, establishing clear performance review processes, championing career development, empowering team members to work autonomously, learning from diverse experiences, supporting team members in navigating power dynamics, having difficult conversations and developing foundational management skills. Active engagement in these areas will help those who lead data science or statistics consulting groups – whether faculty or staff, regardless of title – create and support inclusive teams.

2 | WHO THIS IS FOR

While all team members need to contribute to building an inclusive environment, and leadership can come from people at all levels of an organization, the focus here is on team supervisors, managers or directors ("team leaders"). Whether they have formal managerial responsibilities, team leaders are those who have contact with all team members, influence employment decisions, set policies, establish common practices, undertake performance evaluations and are positioned and empowered to implement the practices discussed here.

To be successful, team leaders need support from not only their team but from their own supervisors and institution executives as well. Those who support team leaders as their supervisors, department chairs, directors or via other senior positions can also implement many of these practices. Going a step further, they can directly acknowledge, promote the impact of and reward this type of work by the team leaders they oversee.

While much of the focus of the literature discussing academic statistics and data science professionals has been on those in faculty-based positions, predominantly ones on the tenure track (Board of Directors of the American Statistical Association, 2018; Mazumdar et al., 2015; Utts & Goretsky, 2021; Waller, 2018), this manuscript is inclusive of all statistics and data sciences consulting teams in academic environments, and potentially in some industrial or governmental environments as well. These are groups that provide statistics and data science services or resources along all parts of the research enterprise, including study design; data collection, management, storage, visualization, analysis; interpretation of results; and grant support. Such groups vary in the research areas they support (e.g. oncology, business informatics, visualization), the units in which they reside (e.g. library, IT, research core, academic department) and the types of positions occupied by team members (e.g. student, staff, tenure-track faculty, research faculty). While faculty in academic departments may engage in some statistics and data science consulting, the teams described in this manuscript explicitly coordinate their activities, share common leadership and self-identify as a team or unit.

3 | ENSURE PEOPLE GET CREDIT

Ensuring that team members receive appropriate credit for their work will support future advancement opportunities

Everyone deserves acknowledgement for their efforts, and it should be an intuitive and straightforward practice. Yet often women, people from marginalized communities, academic staff, students and individuals unfamiliar with merit and acknowledgement practices may find themselves left out without recognition of their efforts and contributions (Liu et al., 2023; Ross et al., 2022). Some cultural differences may exist as well in who gets credit and who does not. Some cultures, for example, may "emphasize social connectedness, interdependence, and in-group goals; while [other cultures may]…emphasize self-reliance, independence, and personal goals" (Cortina et al., 2017). In settings where there is an emphasis on team-based outcomes, team members who are focused on personal goals may have a more difficult time receiving credit than those with in-group goals. Without sufficient credit for efforts, individuals have fewer advancement opportunities, experience low morale and a reduced sense of belonging and may ultimately move on from their position, risking the health of the entire team (Castrillon, 2022; White, 2014). When team members receive credit for their contributions, it not only validates their efforts but also promotes a sense of belonging and equality within the team. This recognition sends a powerful message that every member's unique skills and perspectives are valued, regardless of their background or position within the organization.

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In academia, the primary currency of achievement and advancement is authorship (see Kowalczyk et al., 2022, for example). Yet academic collaborative data scientists and statisticians often have to "be assertive with respect to joint authorship" – something that may be more difficult for those who have been discouraged from such behaviours in the past (Griffith et al., 2022). Therefore, it is important that team leaders support their team members through authorship discussions. Having written policies to which they can refer may set expectations and prevent uncomfortable situations. For example, team leaders can ensure their team members have knowledge of and practice using the broadly relevant International Committee of Medical Journal Editors (ICMJE; n.d.) criteria for defining authorship and contributor roles and the Contributor Roles Taxonomy (n.d.) in discussions about authorship. The ICMJE recommendations include as one sufficient criteria for authorship that an individual provides "substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work." The data science or statistical collaborator often exceeds this criteria by being involved in at least the analysis and interpretation of results. Cooke et al. (2021) provide useful strategies for preventing and navigating authorship disputes that may occur.

While authorship order, frequency of citations and journal impact within the field can play a role in an academician's advancement (see Walker et al., 2010, for example), collaborative statisticians and data scientists should not be expected by their management or position evaluation criteria to be a first, second or corresponding author or publish exclusively in statistics or data science-centric journals (Board of Directors of the American Statistical Association, 2018). Team leaders should applaud successful collaborations that end in publication no matter the author order, the number of citations or the type of journal where publications appear.

Policies for authorship should also address the common assumption that fee-for-service precludes the need for credit in the form of co-authorship. Authorship status is not negated by the payment of fees necessary for a consulting team to operate (LeBlanc et al., 2022). Furthermore, team members in positions that are not evaluated based on publications should nevertheless be credited as co-authors using the same criteria. Forgoing authorship because it is not important for a current position's evaluation can disadvantage people later in their careers. In addition to equity, co-authorship may be important for that individual's professional development should they want to switch institutions, move into a new role in the future or be part of a grant application, they may find themselves in a position where a publication history is expected.

While authorship in academic publications is an important form of credit, it is not the only one. Team leaders should acknowledge the work of individuals in their team through other mechanisms. Some examples may include posting highlights of their work on a website or newsletter, bringing their team member's work to the attention of administrators or suggesting other outlets to disseminate their work. Giving credit means attributing the accomplishments of your team members to them by name, especially to other influential leaders within the institution and the profession. Instead of talking about how successful a specific program has been, team leaders should talk about the success of the named team members collaborating on the program.

Not all credit or recognition needs to come from team leaders or management. Supporting an inclusive environment also means finding ways for team members to recognize and give credit to each other for good work and helping each other. For example, team members can write "high fives" to acknowledge their peers, either on online platforms or by posting notes in a shared office space. This practice lets team members recognize things they value that might go unnoticed by the team leader and helps reinforce team values such as collaboration.

4 | MAKE TACIT KNOWLEDGE EXPLICIT

Documenting and communicating resources and processes leads to a more inclusive and cohesive team

Tacit knowledge is the knowledge gained through experience, usually over long periods of time, and is generally difficult to articulate and share (Polanyi, 1967). Experiences shape the acquisition of tacit knowledge, and variation in these experiences leads to a unique base of knowledge for each team member to draw from. For example, Gardner and Holley (2011) describe challenges that first-generation doctoral students face due to a dearth of tacit knowledge necessary for navigating the academic environment. Surveyed students expressed hardships associated with not knowing which questions to ask and where to seek help. In another example, Carson et al. (2019) describe how early-career faculty from historically underrepresented groups lack the tacit knowledge to navigate service requirements. For example, a statistics or data science consulting team member without an advanced degree may lack the tacit knowledge of academic hierarchies other team members gained through their graduate school experience. Conversely, team members entering the workforce and joining a consulting team straight from their doctoral program may not understand performance review and promotion processes. This "hidden knowledge" can place employees at a disadvantage and have negative implications for retention and functional group dynamics.

For members of data science and statistical consulting teams, this tacit knowledge is most pertinent to understanding how the organization operates. That is, resource navigation, communication norms and standard processes are often implied, and new recruits are expected to "pick things up on their own." To foster an inclusive team, intentional efforts should be made to transfer such tacit knowledge to all team members. In addition to creating inclusivity, this practice can lead to shared learning within the team and improved deliverables and experiences for collaborators.

One approach to foster an equitable working environment in data science and statistical consulting teams is to share knowledge intentionally and transparently, where expectations, processes and evaluation metrics are clearly communicated. Providing team members with explicit

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workflows and a priori expectations of quality serves as a means of transferring the implicit knowledge that senior team members have accumulated through their experiences (Cha et al., 2023). Fostering or explicitly creating peer-mentoring opportunities also promotes knowledge sharing (Nokkala et al., 2022); this can help especially with the transfer of knowledge that team leaders lack, but that can be critical for employees' success. For example, team leaders can create documentation for internal processes, store documents in an easily accessible shared file system and discuss processes openly with all team members regularly to ensure that they are being implemented consistently and that no updates are needed. Sharing resources and tools can also contribute to the teams' tacit understanding and knowledge. For example, team leaders can encourage team members to share and present documentation on common methods (e.g., mixed effects models), tips for using popular tools (e.g., ggplot2) and templates for common email communications and uncomfortable collaboration situations (e.g., rejecting work, referring work to other resources, asking for payment or engaging a collaborator who has lost contact) with the team. This proactive approach can result in increased productivity and fewer mistakes.

5 | ESTABLISH CLEAR PERFORMANCE REVIEW PROCESSES

Clear and explicit expectations and performance review processes enable mutual understanding to the benefit of staff and team leaders

Credit for work and making tacit knowledge explicit become a critical issue when conducting evaluations either through scheduled periodic performance reviews or promotion review processes. Regardless of a data scientist's or statistician's position, the performance review process should provide an opportunity to recognize the impactful contributions that the data scientist and statistician have made for the team, the department and the university (Griffith et al., 2022; Mazumdar et al., 2015). For a team leader, this process is an opportunity to motivate team members, understand their perspectives and needs and recognize achievements. Well-designed, clear and thoughtfully considered evaluation criteria and performance review processes can lead to improved employee engagement, trust between team members and leaders and reduce inequities and unconscious biases for the benefit of overall organizational performance and employee retention (Posselt, 2020, p. 154; Culture Amp, 2023).

Regardless of the individual roles (i.e., staff, faculty) and contract type (i.e, temporary, tenure-track), a transparent performance review process starts by setting defined goals with the team member and clear, corresponding assessment criteria for the team leader to apply. SMART (Specific, Measurable, Achievable, Relevant and Time-bound) goals are one framework that can be used for aligning goals with evaluation criteria (Doran, 1981). Whatever framework is used, both team leaders and team members must have a mutual understanding of the goals and evaluation criteria without ambiguity. For example, a goal or promotion criteria requiring that an individual demonstrate that they are a recognized scholar by publishing in peer-reviewed journals is not sufficient if what is really meant is that the candidate must be the first (or sole) author on a methods development or theoretical statistics manuscript in a top-tier statistics journal. While the appropriateness of this criteria in the modern era of statistics and data science is debatable, if it is a performance requirement, the criteria could explicitly state that the individual should publish 1–3 peer-reviewed journal articles with at least one in a journal from an explicitly named list. As another example, if a data scientist or statistical consultant's job responsibilities specify 50% effort for consultation activities, including hourly meetings, data analysis and grant support, a clear performance review process should indicate whether the consultant has discretion as to which activities they focus on (e.g. allocating all of this effort to consultation meetings and none to grant support) or whether there are expectations for them to meet specific goals in each area (e.g., 4–5 hourly meetings per week). Clearly specifying these criteria will help to ensure that the team member is not spending more of their time in any one, or all, of these components than expected. That is, it is easy to become wholly invested in one component and not meet the needs of the other criteria, or to become burnt out trying to do it all.

Once expectations are set, it is important to track progress. Rather than a one-time, annual meeting to evaluate progress, it may be beneficial for both the team leader and members to schedule periodic meetings to assess and discuss progress towards goals. A bi-directional communication channel is a productive means for building trust and ensuring that the team member has the skills and tools to successfully fulfil their role. An on-going evaluation process helps mitigate problems when they arise and prevent unexpected and possibly unpleasant feedback at a later point. A method that can be employed to provide feedback on performance or to share feedback with the next level of leadership is the STAR (Situation, Task, Action and Result) approach (Development Dimensions International, n.d.). While traditionally used for structuring interview responses, the team leader can use this approach to detail performance examples (situation), describe responsibilities in those examples (task), identify activities relevant to the examples (action) and report on outcomes (result). This format can be used for positive feedback, specifying what was done well or when needed, developmental feedback, which describes a more effective approach, can be provided.

Whereas, for example, the promotion criteria and the eligibility between the faculty levels of Assistant to Associate Professor may be clearly defined, they may be more opaque for promotion for other job roles (e.g., Data Scientist to Senior Data Scientist or Associate to Full Professor). Thus, establishing clear and transparent evaluation criteria is especially important for these team members. For instance, consider the criteria for advancing from a Data Scientist to a Senior Data Scientist. A Senior Data Scientist may be expected to successfully scope and manage projects. The team leader can provide a listing of potential evidence that the Data Scientist can provide to demonstrate they will be successful at the next level (e.g., the type and number of projects that they have contributed to the development of scopes of work).

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An additional area where clarity and transparency are necessary is in the timing of official performance review evaluations, promotion eligibility and the timeline for which particular documents or other evidence are needed. For example, when institutional performance reviews are conducted annually, letting the team member know when these reviews might occur within that year is essential so that the team member can plan accordingly. In addition, providing team members with specific details of the documentation needed, and when it is needed will lead to better preparation and organization. For example, team members could be provided with a template document to complete, or the team leader could share materials from their own performance review or promotion as an exemplar. The timeline for when and mechanisms for how to submit materials can be published on a website so that everyone has the same information (see the Section Make Tacit Knowledge Explicit).

6 | CHAMPION CAREER DEVELOPMENT

Help team members identify and navigate career pathways to retain diverse talent

Data science and statistics are constantly evolving fields. For team members to maintain their knowledge and expertise, ongoing technical learning and training is required. Supporting career development begins by allocating time and resources for team members to engage in this foundational professional growth and learning.

However, championing career development requires more than sending team members to conferences or classes. Team leaders should help team members think about their whole career trajectory – possibly beyond their current position or organization. This forward-thinking necessitates honest and clear evaluations and discussions of a team member's current skillset, the skills needed for the next step in their career ladder, the skills that they would like to develop and avenues for developing the skills (see Van Tuyl, 2023, for examples of academic data scientist career ladders). Team leaders can help identify those avenues by mapping current work and accomplishments to skills team members are looking to develop or need for their next career stage. While team leaders can help, ultimately, team members need to take ownership of their own career development. Two steps team leaders can encourage team members to take are connecting with others who are already in the types of positions they would like to move into and looking at the skills required in job postings of positions that interest them. Once there is a joint vision of the skills and experience team members would like to achieve, team leaders can seek opportunities for team members to engage in that type of work.

By seeking out opportunities that align with their team members' goals and using their position to advocate for their team members' ability to take on new challenges, team leaders can become a 'sponsor' and not just a 'mentor' (Hewlett, 2013). A sponsor is a professional champion that "guides others toward [the sponsee's] work and contributions" (Ensor, 2015) and goes beyond advice to support and advocacy, "connecting [the sponsee] to important players and assignments" (Hewlett, 2013). Sponsorship has been shown to improve retention and advancement for individuals from underrepresented groups (Hewlett & Ihezie, 2022).

Sponsorship ideally involves identifying significant opportunities for team members to stretch their abilities and advance their careers. However, team leaders should also provide more routine pathways (e.g., time, resources) for individuals to participate in activities and create deliverables needed for advancement. For example, in a data science collaboration context, suppose that a key evaluation criteria for an individual is to present a collaborative research project at a conference. The team leader can work with the team members on time allocation (e.g., adjusting time spent on other projects as necessary) and can provide resources (e.g., funding) to present the work. Less directly, team members can present their work regularly within the group to support individual presentation and communication skills.

7 | EMPOWER TEAM MEMBERS TO WORK AUTONOMOUSLY

Fostering independence communicates the value of team members' work and supports a culture of contribution

Empowering team members and fostering autonomy and independence in employees is crucial for promoting creativity, innovation and personal growth, as well as the retention of talent (Cagan & Jones, 2020). Empowerment means creating an environment where people have ownership over the outcomes of their work rather than only the tasks (Cagan & Jones, 2020). Team leaders can value, reward and celebrate independent thinking and initiative to reinforce the autonomy and independence of all team members.

An important aspect of empowering team members is the delegation of authority to give people ownership of their work. Delegation allows decisions to be made by those with subject expertise and understanding of the issues, who are most likely to be the first to recognize appropriate solutions (Bloom, 2020). When delegating, a team leader must be willing to take the risk associated with letting someone else be responsible for a project, trust team members to successfully complete the task and be patient while they may be learning new skills (Nelson, 1993; see, for example, the "independent R&D" of Nowviskie, 2013). For the delegation to be successful, it is imperative that team leaders clearly communicate expectations, goals and the details of authority granted for the completion of the project. Team leaders can then adopt a coaching approach (i.e., working alongside people rather than just guiding them) and help team members utilize their potential without asserting control over the project (i.e., micromanaging) or taking credit for their accomplishments (Cagan & Jones, 2020). These approaches require active work from the team leader to assess their team, develop coaching plans and offer regular feedback and guidance. As a part of coaching, the team leader can share their

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own experiences with balancing projects, having tough conversations with clients or setting professional goals. Inviting team members to attend meetings with stakeholders may also be beneficial to both the manager and members of the team. In addition to receiving the most current information about the project's status (team members may be more familiar with the project's progress than the team leader), partnering with team members in this way builds trust, communicates appreciation and inclusion and may serve as a professional growth opportunity for the team member.

8 | LEARN FROM DIVERSE EXPERIENCES

Create space and opportunities for team members to share how they experience the world and how that affects their work

Facilitating conversations with and among team members is important for reaping the benefits of a diverse team and creating a culture that is inclusive of varied experiences. As Rock et al. (2016) put it, "capitalizing on diversity means highlighting — not hiding from — differences." Similarly, White's (2022) suggestions for leaders looking to build an inclusive organizational culture start with "actively listening and learning" about the experiences of team members. Yet 61% of employees report "covering," or taking steps to avoid drawing attention to, at least one aspect of their identity that may have a stigma associated with while at work, even in workplaces with stated commitments to inclusion (Smith & Yoshino, 2019).

Successfully navigating complex conversations about personal differences requires building shared values and trust within a team. Pérez (2021) outlines steps to build trust through conversation and help a team progress to having transformative conversations in emotionally intelligent ways: 1) asking discovery questions, 2) actively listening, 3) using dramatic emphasis and 4) acknowledging life milestones and work achievements. Su (2019) suggests steps for building trust and psychological safety in a team, including modelling healthy conflict, zero tolerance for bullying and developing a culture of appreciation. It is critical to address bullying and poor interpersonal behaviours with care and immediately, as team members' work and development can be significantly impeded if not.

Including someone as part of a team means listening to and learning from their story. Each team member has a unique history, identity and personality that will affect their experience within the team, in their consultations and within the institution. These personal identities and histories (e.g., age, gender, race and more) will affect their experience. For data science and statistics consulting teams, there are also varied experiences in when and how people learned to code, familiarity with different research domains, level of degree, role at the university (faculty, staff, student) and job title. Beyond their backgrounds, team members will also have differences in how they experience ongoing events, such as changes in the work environment and broader news and political events.

When learning about the experiences of others, avoid either reducing their experience to one aspect of their identity or ignoring the systemic factors that may contribute to their experience. Team leaders can also listen to how team members frame their experiences and share their own understanding rather than interpreting it for them. For example, both responding to a particular woman's experience with discrimination by saying it's something all women face and saying that discrimination is something many people of different identities face can be dismissive of someone's individual experience. The former may deny the particular ways an individual's multiple identities of race, sexuality and age intersect with gender in their experience. The latter ignores the specific systemic factors that disadvantage women, which can undermine efforts to address them.

With data science and statistics consulting teams, learning from diverse experiences should also extend to exploring how collaborators interacting with the team may have different experiences. For example, raising awareness of how a first-generation undergraduate student's experience consulting the team may differ from that of an early career faculty member with a history of qualitative research will help team members be more effective in their collaborations. Being intentional and direct in discussing how individuals' differences affect their experience working with the team will also help team members reach out to new collaborators in ways that speak to their needs and experiences.

9 | SUPPORT TEAM MEMBERS IN NAVIGATING POWER DYNAMICS

Position, identity, situation and expertise can all affect the power dynamics between those involved in collaboration and contribute to or hamper the success of the interaction

Identifying and navigating power dynamics is necessary on both sides of a consultant/client relationship. As with learning from diverse experiences, acknowledging and explicitly having conversations about the factors that shape academic consulting relationships is critical for addressing issues that can inhibit an inclusive environment.

The power relations or dynamics (see Foucault, 1980; Stainback et al., 2010) between a consultant and a client will be affected by

- Position: the role in the organization that both people occupy (e.g., undergraduate student, graduate student, staff, faculty, IT support, researcher staff).
- Identity: the personal characteristics and identities of each, including age, gender, ethnicity/race, abilities/disabilities, etc.

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- Situation: the parameters under which the two parties are interacting, such as the definition of the service being offered, the expectations for collaboration and any financial exchange involved.
- Expertise: the expertise that both parties bring to the content of the interaction, as well as their overall expertise levels in related areas.

Position is mostly a function of the institution and system that people are operating in and is not easily changed. Supporting team members on this dimension starts with making any tacit knowledge about assumed status differences between role types explicit and acknowledging it. For example, acknowledging that faculty consultants on a team may be less likely than staff consultants to need to advocate for their role as a co-author on a paper. Positional power differentials can be addressed by adopting practices and norms for the team that attempt to adjust the dynamic. For example, the team can use first names as the default form of address for both team members and clients. Consulting appointments could be scheduled in the consulting team's space rather than a consultant being expected to travel to a faculty member's office. Team members can include academic credentials or positions in their email signatures.

While *identity* is also based on characteristics that are largely fixed, the ways in which individuals' identities interact depend upon cultural and social context. For example, being Sikh in India carries with it a complex cultural and political context of which people in the US may be completely unaware. In the multicultural environment of a university or research institution, people bring their cultural and social backgrounds, including their expectations for interacting with people of different identities and assumptions about power dynamics. Supporting team members in educating themselves about such differences through professional development funding or goals is one way to improve their ability to navigate these dynamics.

Power dynamics that result from the *situation* are the ones that can most easily be influenced by the team leader. As Pozzebon and Pinsonneault (2012) found, the initial set-up of a project is particularly important. Having clear in-take or collaboration initiation processes that reference written policies can help to set expectations up front and frame the interaction. Written policies also make it easier for consultants to address breaches in the expectations or scope of the relationship by providing something they can reference. For example, written project agreements that clearly outline the scope of the engagement can help consultants address requests that would expand the scope – such as investigating new questions or engaging in data cleaning tasks that were not part of the initial agreement.

Power differentials resulting from *expertise* may help a consultant who may otherwise feel they are in a weaker or vulnerable position in a consulting engagement. However, a consultant needs to be confident in their expertise and see themselves as an expert for their expertise to play an equalizing role in a relationship, as well as being comfortable in unknown and new areas. Consultants who have received training are anecdotally more likely to develop this confidence (Sharp et al., 2021). Similarly, a student seeking data science assistance from a faculty member may gain confidence when reminded that they are the experts in their own research domain and project. Asking someone seeking assistance to describe their research or provide input on how they have seen a problem addressed in other research in their field can help them access and share this expertise, leading them to be more comfortable sharing other areas where they need assistance.

Helping consultants recognize these dynamics and explicitly discussing strategies for addressing them, both individually and as a team, can help the team have the language and a framework for discussing issues that may arise in their consulting engagements. Sharp et al. (2021) developed 10 statistical collaboration training videos and scripts along with discussion questions as a resource for facilitating these discussions.

10 | HAVE THE DIFFICULT CONVERSATIONS

Advocating for the team requires sometimes instigating and engaging in uncomfortable or difficult conversations with team members, collaborators or unit leadership

Being an inclusive leader requires courage (Bourke, 2016). While this may come in many forms, one place courage is required is having conversations that address when a team member, collaborator or senior leader is failing to meet your team's expectations for an inclusive environment.

With team members, this can mean enforcing zero-tolerance policies for bullying, racist or sexist language or other biased or exclusionary behaviours. This may involve challenging team members who use potentially harmful language that could be derogatory to consider the history and implications of their word choices. It may require reminding team members that their peers or collaborators may have family members who are affected by government policies or actions they are vocally supporting or opposing. Importantly, team leaders should recognize these behaviours in action and acknowledge the personal experiences of those that report them. Specific to data science or statistics projects, the team leader may need to remind or educate team members of the complex history and implications of data encoding choices around race, ethnicity, citizenship, gender and other personal characteristics. It takes courage to directly and clearly address such situations, but doing so is essential for setting clear expectations for the team.

With collaborators, team leaders need to proactively ask about team members' interactions with constituents or clients and be prepared to engage in conversations with those collaborators to address any issues. For example, a data scientist working with a faculty member who repeatedly uses inappropriate language may be hesitant to bring this up, let alone address it themselves, especially given the power differentials

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involved. Similarly, a statistician may be working with a researcher who refuses to list them as a co-author, even though the statistician wrote the methods section of a paper. Supporting team members in these situations means setting expectations that such behaviour is unacceptable, that team members should come to the team leader if something like this occurs, and that the team leader will take action if team members do report it. Unless these expectations are explicitly stated or team members observe the team leader engage in such conversations, team members may assume they have to deal with such situations on their own.

Team leaders may also need to have difficult conversations with their own managers or stakeholders. As with team members or collaborators, team leaders may need to address behaviour by senior leaders that hinders the inclusive environment they are working to lead. However, difficult conversations may also extend to ways in which the systems or policies in place are failing to support the inclusive environment the team leader is working hard to build and support. For example, units that recruit job applicants through personal networks instead of open job postings, use exclusionary language in job descriptions, exclude applicants based on poor "culture fit" or discount applicants that have any gaps in their work history are undermining efforts to support an inclusive environment from the start. Conversations to change such practices are likely to be difficult, but having them is critical.

Difficult conversations do not need to be punitive. Aggressive or harmful behaviours or words may not be intentional, but left unaddressed, harmful dynamics are likely to continue. Approaching difficult conversations collaboratively can help everyone move towards more productive and inclusive relationships. The goals are to set clear expectations, establish that people will be held to the established standards and create an environment for improved collaboration moving forward. See Patterson et al. (2002) and Hancher (2022) for tips on having these difficult, but critical, conversations.

11 | DEVELOP MANAGEMENT SKILLS

Developing skills to consistently and equitably implement best management practices is a prerequisite for creating an inclusive environment

Diverse teams have many benefits, but they also create challenges that must be managed for the benefits to be realized (Rock et al., 2016). The practices discussed in earlier sections become easier to enact if team leaders have good fundamental management practices in place that help establish positive and trusting relationships with team members.

In our experience, data science or statistical consultant team leaders, like other academic leaders, rarely receive training in how to be an emotionally intelligent team leader as part of their academic education or career. Fortunately, there are many good resources on how to build these crucial skills, likely including training courses and programs offered by their institutions. Seeking out and participating in such training is not a nice-to-have or bonus activity – it is core to the work of being a team leader and needs to be prioritized as such. From email newsletters to books, from podcasts to online courses to multi-day training, there are multiple ways to engage in continuous learning (see Table 1 for some places to start).

Foundational management practices include: holding regular one-on-one meetings with members of your team - including students, providing both positive and constructive feedback on a regular basis, establishing regular methods and practices for communicating with the team and having explicit discussions about job performance expectations. Performing management tasks well requires building trust with the team and employing emotional intelligence, both of which can be built through specific practice (Pérez, 2021; Sebre et al., 2021).

Managing up is also a crucial skill (Rousmaniere, 2015). Supporting the statistics and data science team through the practices above will only go so far if the next-level team leader, department chair or stakeholders are not supportive partners in the process. The statistics and data science

TABLE 1 Resources to start further learning.

Management skills
Manager, Ph.D. email newsletter: https://www.managerphd.com
Manager tools podcasts: https://www.manager-tools.com/all-podcasts
The effective manager, by mark Horstman et al.
"Hiring, Managing, and Retaining Data Scientists and Research Software Engineers in Academia: A Career Guidebook from ADSA and US-RSE," https://zenodo.org/doi/10.5281/zenodo.8264152
LinkedIn learning: https://www.linkedin.com/learning check if your institution provides access
Allyship and inclusive leadership
"The EThIC Model of Virtue-Based Allyship Development: A New Approach to Equity and Inclusion in Organizations," Warren & Warren, 2021, *Journal of Business Ethics*, 10.1007/s10551-021-05002-z
"The six signature traits of inclusive leadership," Juliet Bourke, 2016, https://www2.deloitte.com/us/en/insights/topics/talent/six-signature-traits-of-inclusive-leadership.html
Anti-racist leadership, by James D. White and Krista White
Leadership in statistics and data science: Planning for inclusive excellence, edited by Amanda L. Golbeck
Empowered: Ordinary People, Extraordinary Products, Marty Cagan and Chris Jones, 2020
Employee Empowerment: The Prime Component of Sustainable Change Management, Daniel Bloom, 2021

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TABLE 2 Summary of leadership and management practices discussed in this work.

Practice	Description	Examples
Ensure people get credit	Ensuring that team members receive appropriate credit for their work will support future advancement opportunities.	Advocate for authorship on publications. Publicly recognize team members' contributions regardless of authorship status. Provide opportunities for team members to recognize each other's work and contributions to the team.
Make tacit knowledge explicit	Documenting and communicating resources and processes leads to a more inclusive and cohesive team.	Document workflows and standard operating procedures within the team. Create mentoring opportunities. Internally share resources and tools, such as documentation on common methods (e.g., mixed effects models), tips for using popular tools (e.g., ggplot2) and templates.
Establish clear performance review processes	Clear and explicit expectations and performance review processes enable mutual understanding to the benefit of staff and team leaders.	Put evaluation criteria in writing. Discuss evaluation criteria with team members to ensure shared understanding. Regularly discuss performance progress and challenges with team members.
Champion career development	Help team members identify and navigate career pathways to retain diverse talent.	Discuss professional aspirations with team members. Help team members identify connections between their work and the skills needed for the next step in their career. Sponsor team members in pursuing activities and opportunities that will grow their experience and skills.
Empower team members to work autonomously	Fostering independence communicates the value of team members' work and supports a culture of contribution.	Delegate responsibility. Coach team members, do not micromanage.
Learn from diverse experiences	Create space and opportunities for team members to share how they experience the world and how that affects their work.	Listen. Avoid interpreting other people's experiences and identities for them. Consider the experiences of both team members and those collaborating with the team.
Support team members in navigating power dynamics	Position, identity, situation and expertise can all affect the power dynamics between those involved in collaboration and contribute to or hamper the success of the interaction.	Explicitly discuss strategies for navigating status differences, such as norms around the use of titles and being explicit about a project's scope. Support team members in learning about ways different identities may influence people's experiences.
Have the difficult conversations	Advocating for the team requires sometimes instigating and engaging in uncomfortable or difficult conversations with team members, collaborators or unit leadership.	Address inappropriate language by collaborators, especially when directed at team members. "Manage up" to change exclusionary language in job descriptions.
Develop management skills	Developing skills to consistently and equitably implement best management practices is a prerequisite for creating an inclusive environment.	Pursue training in leadership and management skills. Hold one-on-one meetings with all team members regularly.

team leader can effectively manage by holding regular meetings with their leadership, establishing channels for regular and open communication, seeking joint professional development opportunities and educating their supervisor about what they need to be effective in their position as a team leader.

Once the basics are in place, team leaders should think about how to develop team members' management and leadership skills (see MacGillivray, 2021 for suggestions on diverse ways to develop leadership skills). Team leaders can create an explicit mentorship program or provide opportunities for the team to learn and practice these skills. Opportunities that exist for the statistics and data science team leader might also be relevant to team members, either separately or concurrently, as well.

12 | CONCLUSION

This manuscript provides ideas for ways in which team leaders can do good and lead an inclusive statistics or data science consulting team (Table 2). Creating an inclusive and collaborative working environment may be time-consuming initially. However, this should be considered a

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long-term investment that will lead to increased team members' engagement and creativity. Team leaders do not need to apply everything discussed in order to have an inclusive team. The goal for team leaders at all stages of their journey to better support their teams is to identify one specific action they can take next and continue to implement. Team leaders who are new to management or have yet to establish good management practices should start in the last section to learn how to build a foundation of communication and trust with their team. Those who have an established foundation can use the ideas presented to identify and explore new practices that are relevant to their team and are consistently achievable; in the spirit of the overall goal, team leaders are encouraged to ask team members to anonymously identify areas where they would like to see improvement. Team leaders who have been doing this work for years have the opportunity to mentor and inspire future leaders as well as share their experiences that can positively shape statistics or data science teams (e.g., through manuscripts or presentations). The statistics and data science community needs to hear from more people who prioritize leading inclusive teams in order to establish shared expectations for these practices throughout the field.

AUTHOR CONTRIBUTIONS

Christina Maimone, Julia L. Sharp, Ofira Schwartz-Soicher, Jeffrey C. Oliver, Lencia Beltran: Conceptualization; writing—original draft; writing—review and editing.

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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